ASSIGNMENT NO 1
GSC-101

Question-1(4 Marks)

Write down two real life examples where object is moving with positive velocity but having negative acceleration?

1. When you step on the breaks of your car you have positive velocity and negative acceleration assuming that you coordinate system is situated such that the direction you are traveling is positive.
2. A stone thrown straight up. As long as it is climbing up, it has positive velocity and negative acceleration

Question-2(6 Marks)

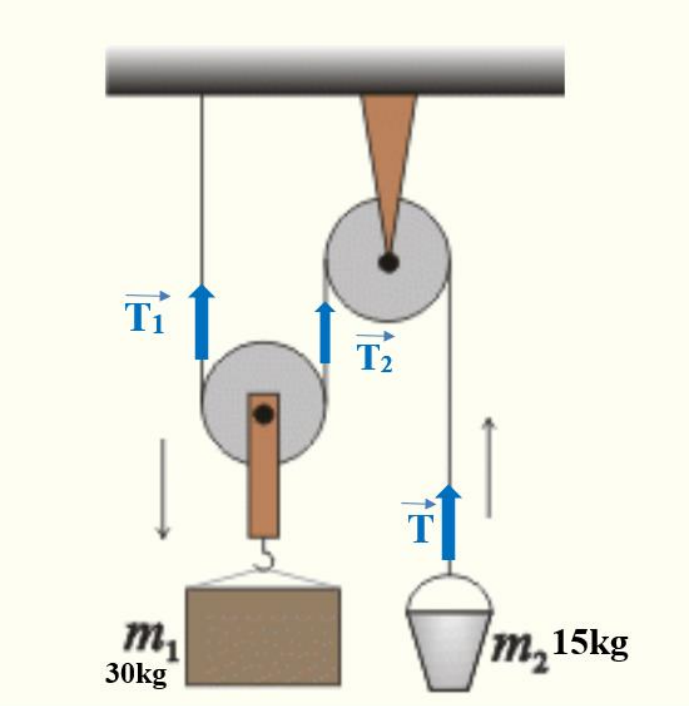
Read the following statements carefully, discussed which Newton law of motion is applicable and also explain reason.

Statement	Newton Law of Motion	Explanation according to Newton
1. If you release a balloon in air	**Newton 3rd law of motion**	According to Newton 3rd law of motion action and reaction are equal but in opposite direction. The action is by the balloon that pushes the air out of it when set free. The reaction of the air which escapes out from the balloon acts on the

		balloon. It is due to this reaction of the escaping air that moves the balloon forward
2. You need more force to move and accelerate a heavier objects than lighter objects.	**Newton 2nd law of motion**	According to Newton 2nd law of motion acceleration is directly proportion to the net force. Therefore, we need more force to move and accelerate a heavier object than lighter objects
3.A child riding a merry-go-round lets go and is thrown off the ride	**Newton's 1st law motion**	The important part of Newton's 1st law motion to recall is that objects will remain in motion in the same direction. The rotational motion of child will result in the constantly changing velocity in the direction tangent to the edge of the ride

Question-3(10 Marks)

In the given figure two objects are hung on a frictionless pulley. The block of mass (m_1) moves down and the bucket of mass (m_2) moves up. Find out the magnitude of acceleration(a)of these two objects and tension (T) in the string. (Neglect the masses of pulley and rope)



Forces that affect the bucket m_2:

W_2 = Weight

T = Tension force by which the rope affects the bucket

Forces that affect the block m_1

W_1 = weight

T_1 = Tension force by which the rope affect the block

T_2 = Tension force by which the rope affect the block

For bucket the equation to scalar form

$T - W_2 = m_2 a_2 \longrightarrow$ 1

$W_1 - T_1 = m_1 a_1 \longrightarrow$ 2

Because we leave the mass of the pulley system and the rope out of the account, they have no moment of inertia and don't affect the tension the forces. The following hold for the magnitude of the tension forces.

$T = T_2 = T_2$

We write the above equation

$T - W_2 = m_2 a_2 \longrightarrow$ 3

W_1-2T = m_2a_1 ⟶ 4

Relation between the magnitudes of the acceleration a_1 and a_2 of the tension forces.

If the bucket goes up a distance S, the block goes down a distance S/2

$S=\frac{1}{2}a_2t^2$

$\frac{S}{2}=\frac{1}{2}a_1t^2$

By above two equation we get

$2=\frac{a_2}{a_1}$

$2a_1= a_2$ ⟶ 5

So, the equation (3) and equation (4) can be written as

$T-m_2g = m_2 2a_1$ ⟶ 6

$m_1g - 2T = m_1 a_1$ ⟶ 7

We have two equation (6) and (7) in two variables T and a_l.

From them we can determine the magnitude of the acceleration a_1 and the force T.

We multiply equation (6) by 2 and both equations up:

$m_1g + 2m_2g= m_1a_1 + 4m_2a_1$

$(m_1 +2m_2)g = (m_1 + 4m_2)a_1$

$m_1g-2T = m_1a_1$

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$m_1g+ 2m_2g = m_1a_1+ 4m_2a_1$

$(m_1 +2m_2)g = (m_1 + 4m_2)a_1$

$a_1=\frac{(m_1+2m_2)g}{(m_1+4m_2)}$ ⟶ 8

According to equation (5) becomes

$a_2=2\frac{(m_1+2m_2)g}{(m_1+4m_2)}$ ⟶ 9

The magnitude of the tension force T is given for example by the equation (6)

$T-m_2g = m_2 2a_1$

Z.F

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